

Constant-Round MPC Protocols with Fall-Back Security

TCC 2026

Anasuya Acharya
Aalborg University

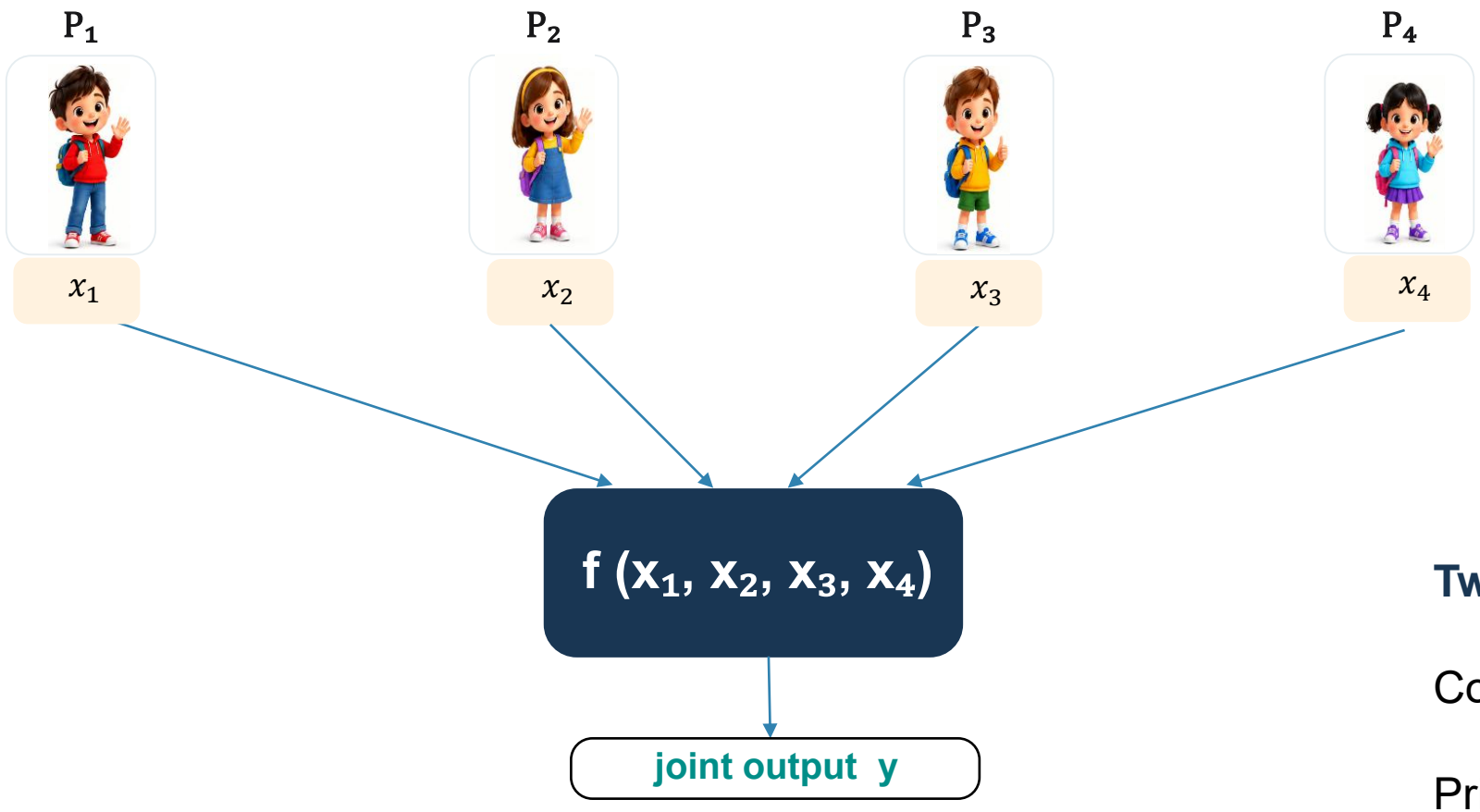
Aditya Patankar

Arpita Patra
IISc

Divya Ravi
University of Amsterdam

Raghavendra Vernekar
IISc

Secure Multiparty Computation (MPC)

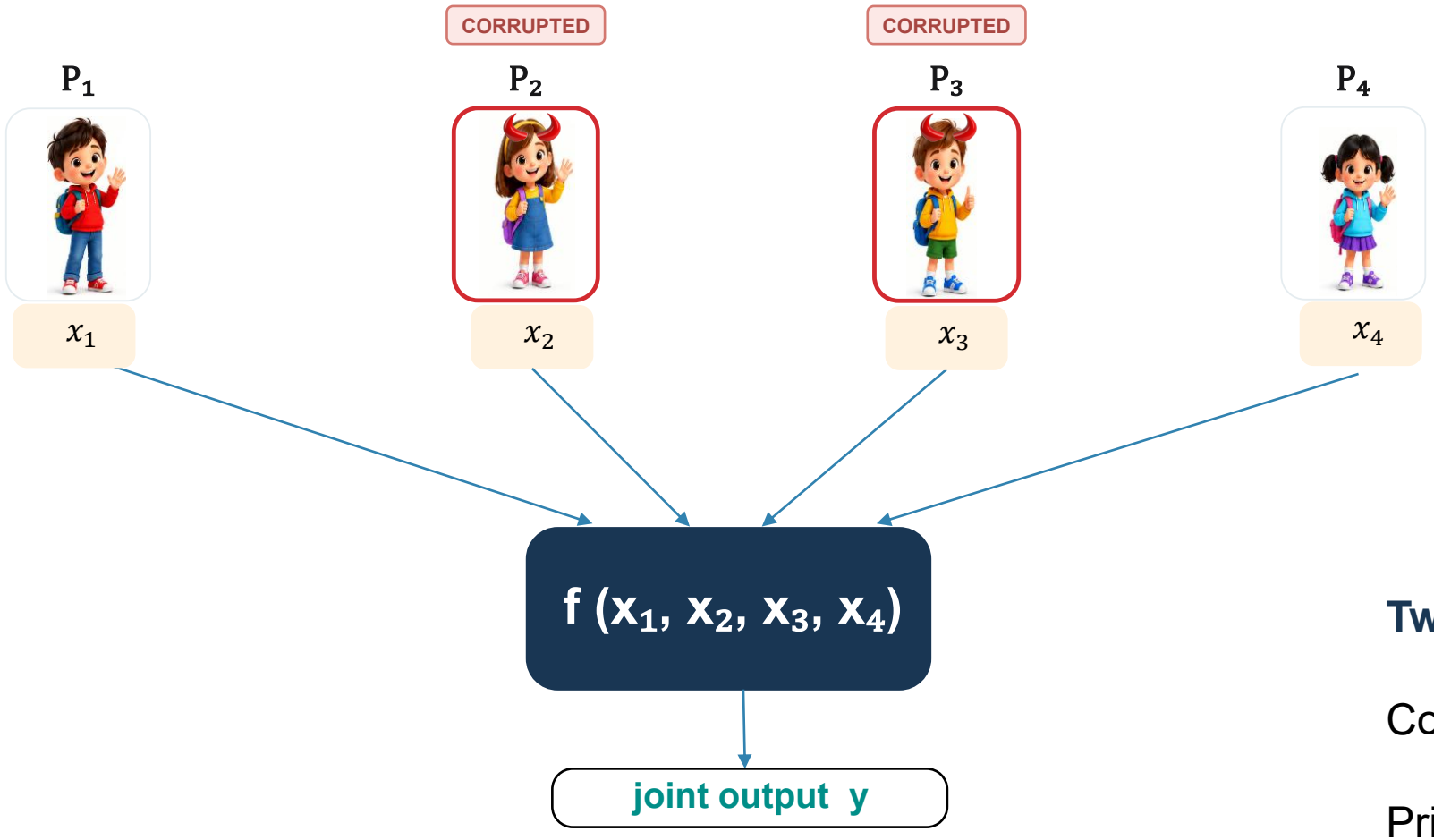


Two Properties

Correctness of Output

Privacy of Inputs

What If Some Parties Are Corrupted?



Two Properties

Correctness of Output

Privacy of Inputs

P_2 and P_3 are corrupted; security must still protect the honest parties.

Adversarial Behavior



Semi-honest

Follows protocol

Runs the prescribed computation but examines its view to infer private information.

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Runs the prescribed computation but examines its view to infer private information.



Malicious

Can deviate arbitrarily

May send incorrect messages, omit broadcasts, or force an abort; additional checks are needed.

How Powerful Is the Adversary?



Unbounded adversary

Does not rely on any hardness assumption

Information-Theoretic Security

for t < n/2

t = number of corrupted parties; n = total number of parties

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PPT adversary

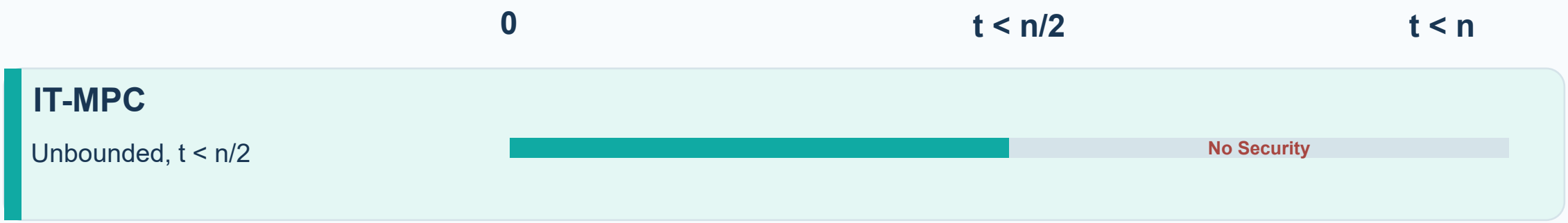
Security relies on computational hardness

Computational Security

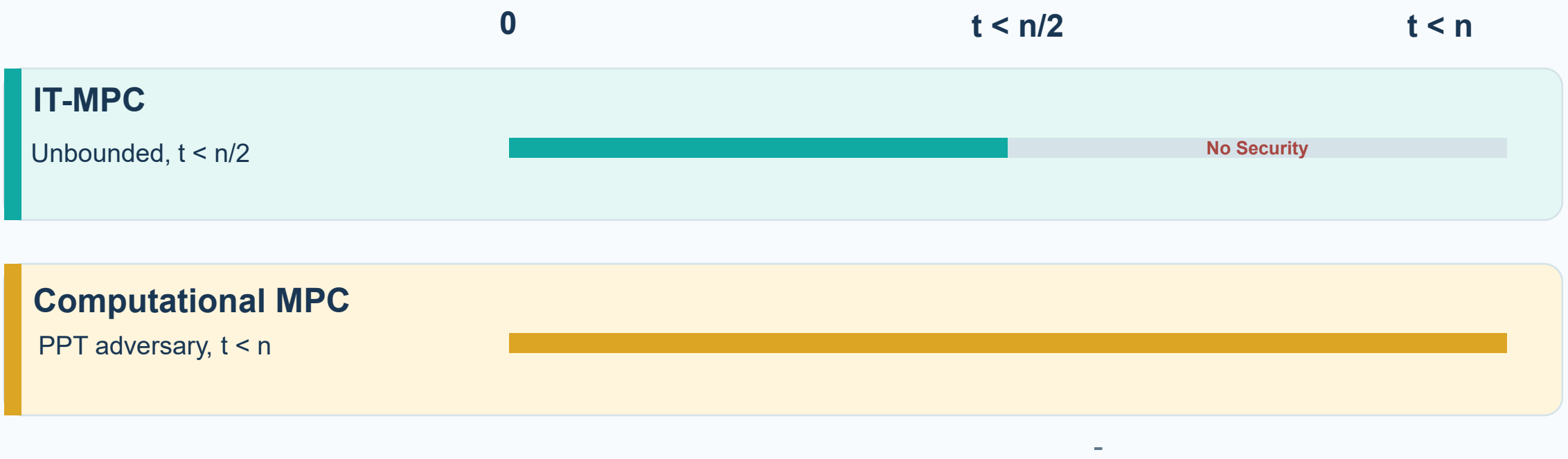
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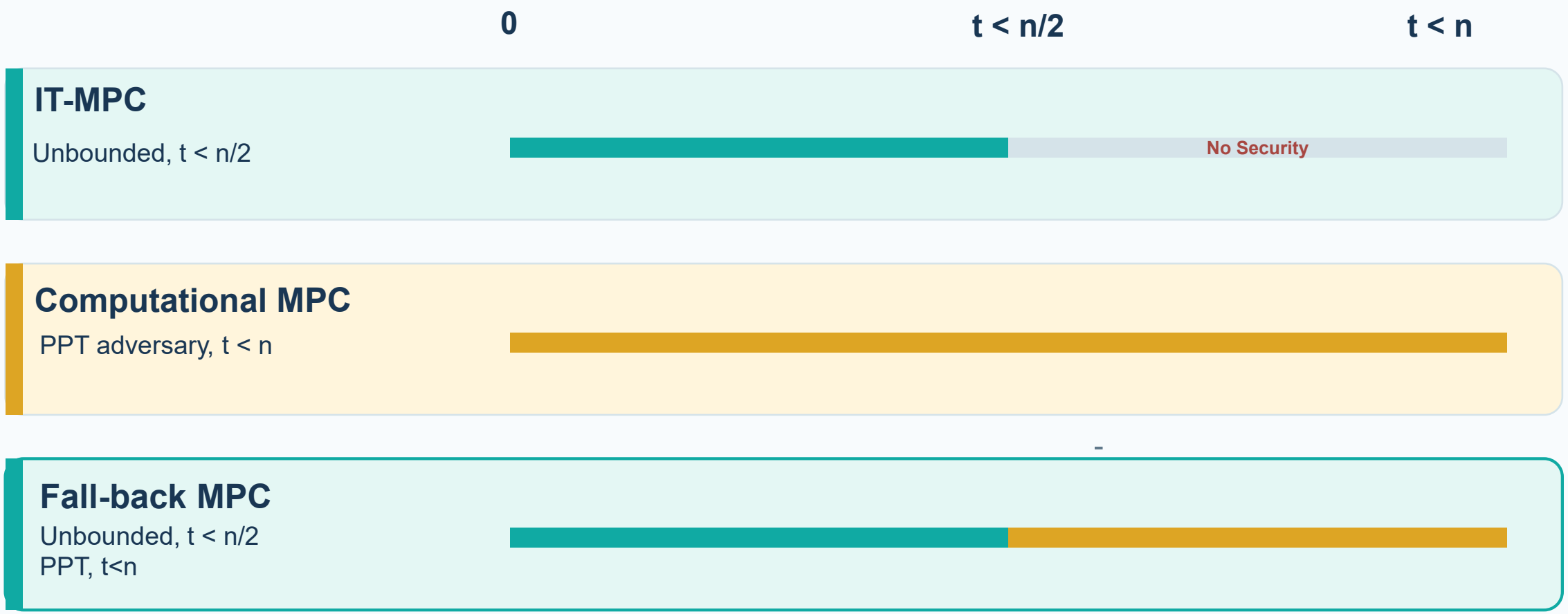
Fall-back Security: Best of Both Worlds



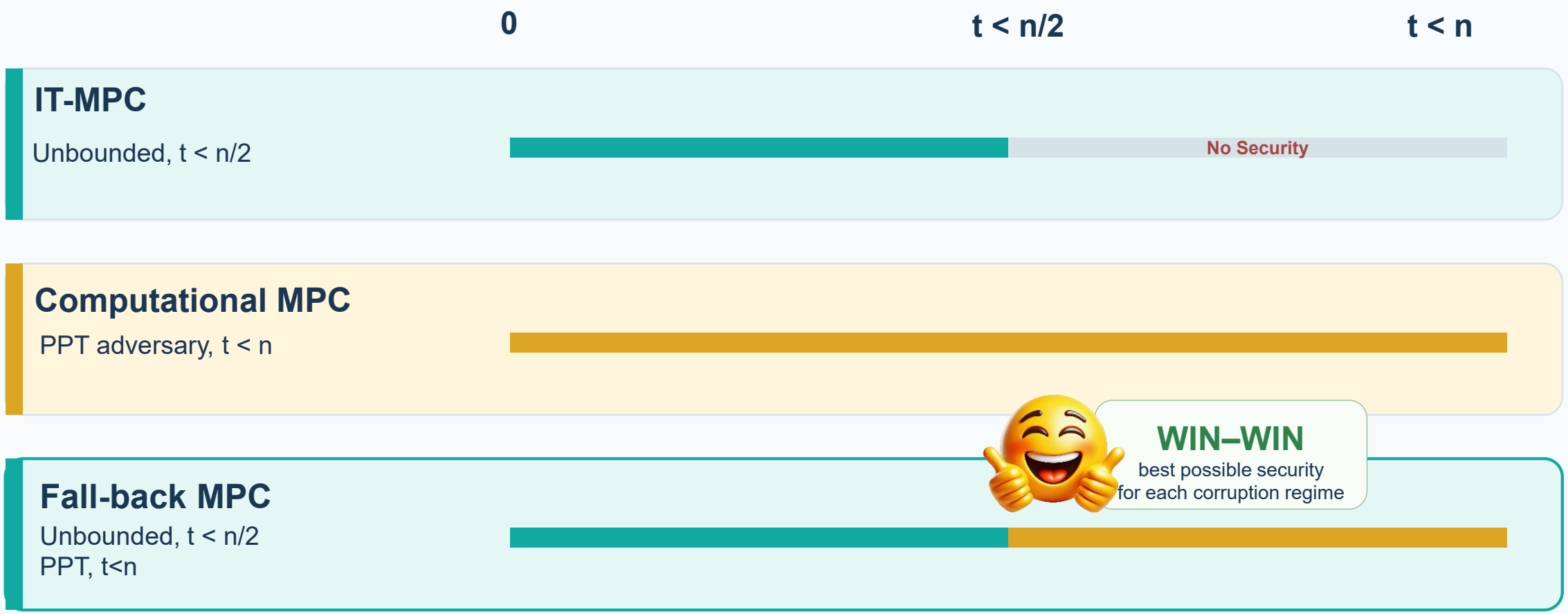
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From Feasibility to Round Efficiency

11

Semi-honest fall-back MPC

Acharya et al. (CRYPTO 2023): at least 11 rounds.

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OUR CENTRAL QUESTION

Can we achieve fall-back-secure MPC in a constant (small) number of rounds?

Our Results: Constant-Round Constructions

3

Semi-honest · Plain model

Statistical: unbounded $t < n/2$ | **Computational:** PPT $t < n$

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Malicious · Fairness + UA

Fairness: unbounded $t < n/2$ | **UA:** PPT $t < n$ · CRS model

Designated-Party Protocol: Two Security Regimes

Fix a designated evaluator P_i. The guarantee depends on the adversary.

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If P_i is corrupt and the adversary is unbounded, the protocol guarantees no security

Key Idea: Emulating Virtual Parties

Start with an IT-secure virtual MPC; emulate its parties securely.

VIRTUAL PROTOCOL

IT-secure MPC

V_1 V_2 V_3 $\cdots$ V_n

Each virtual party V_i has its own state.

EMULATE

REAL-PARTY EXECUTION

Designated-party protocols

$P_1 \rightarrow V_1$ $P_2 \rightarrow V_2$ $P_n \rightarrow V_n$

P_i is the designated evaluator for V_i.

From 2-Round MPC to 3-Round Fall-Back MPC

Starting point: a 2-round semi-honest IT-secure MPC protocol for $t < n/2$.

INPUT · VIRTUAL PROTOCOL

2

rounds

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Honest majority: $t < n/2$

ABT18 · Perfect Secure Computation in Two Rounds

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OUR COMPILER

Emulate virtual parties via designated protocols

Prepare output garbling in parallel



OUTPUT · FALL-BACK MPC

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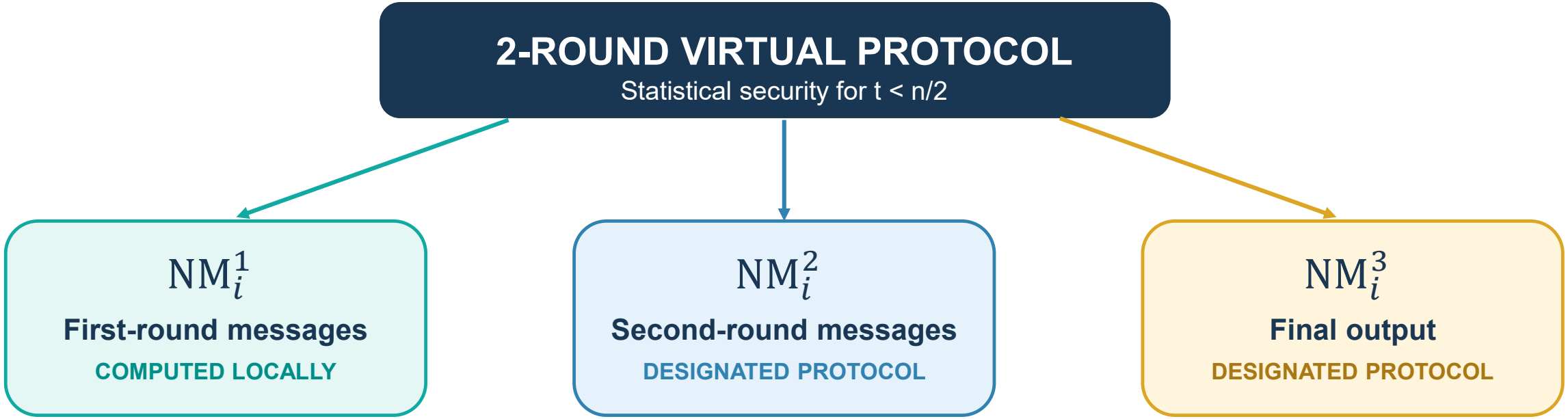
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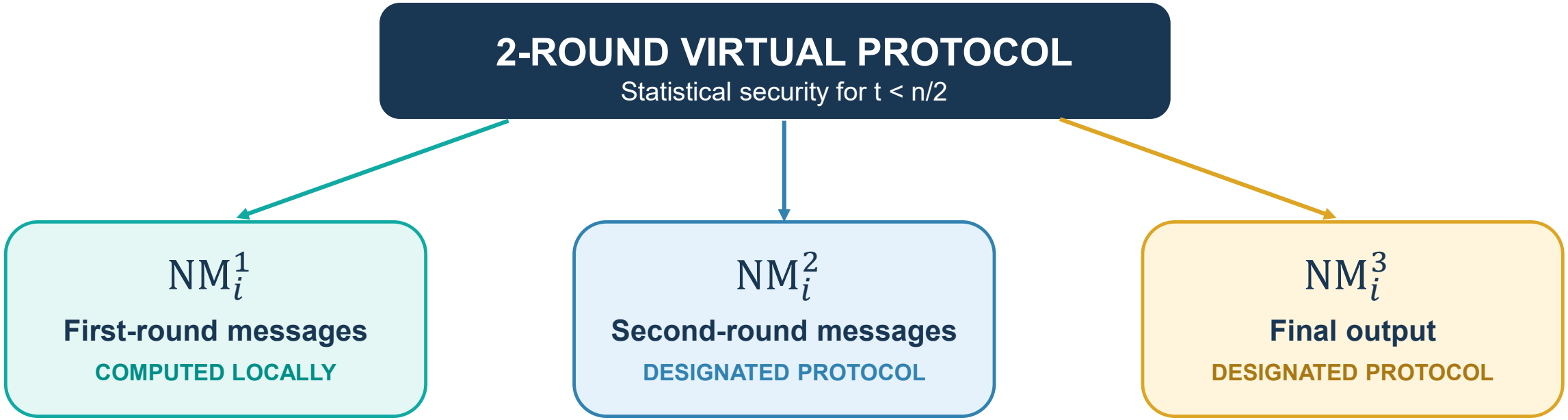
The Virtual Protocol

Represent each virtual party V_i by its next-message functions.



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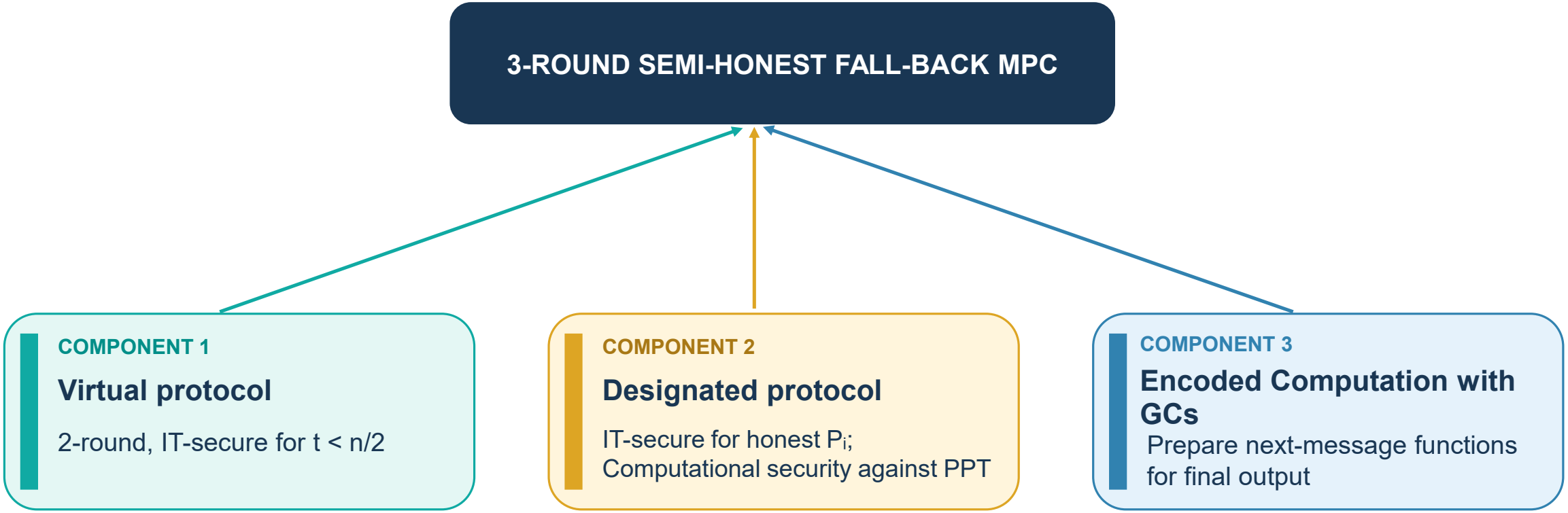
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NM_i^1 is computed locally; NM_i^2 is computed and the garbling of NM_i^3 is prepared in parallel using designated-party protocols.

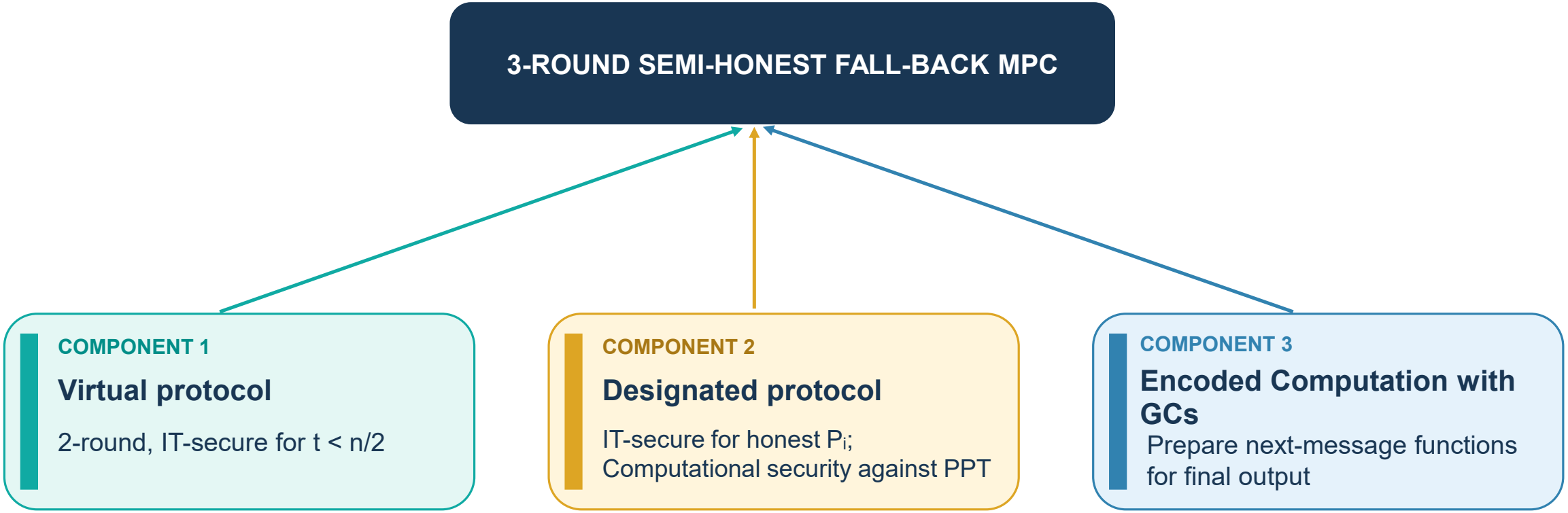
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Key optimization: compute NM_i^2 and prepare NM_i^3 in parallel → 3 communication rounds.

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Honest majority (t < n/2)

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IT security reduces to the virtual protocol; PPT security relies on designated protocol + encoded labels.

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These checks let honest parties reach the same abort decision.

Conclusion: 3, 4, and 5 Rounds

Protocol	Rounds	Honest majority	Dishonest majority
Semi-honest	**3**	IT security	Computational security
Malicious UA	**4**	IT security + UA	Computational UA
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All results: NC^1 functionalities · Secure point-to-point + broadcast channels

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Thank You!

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